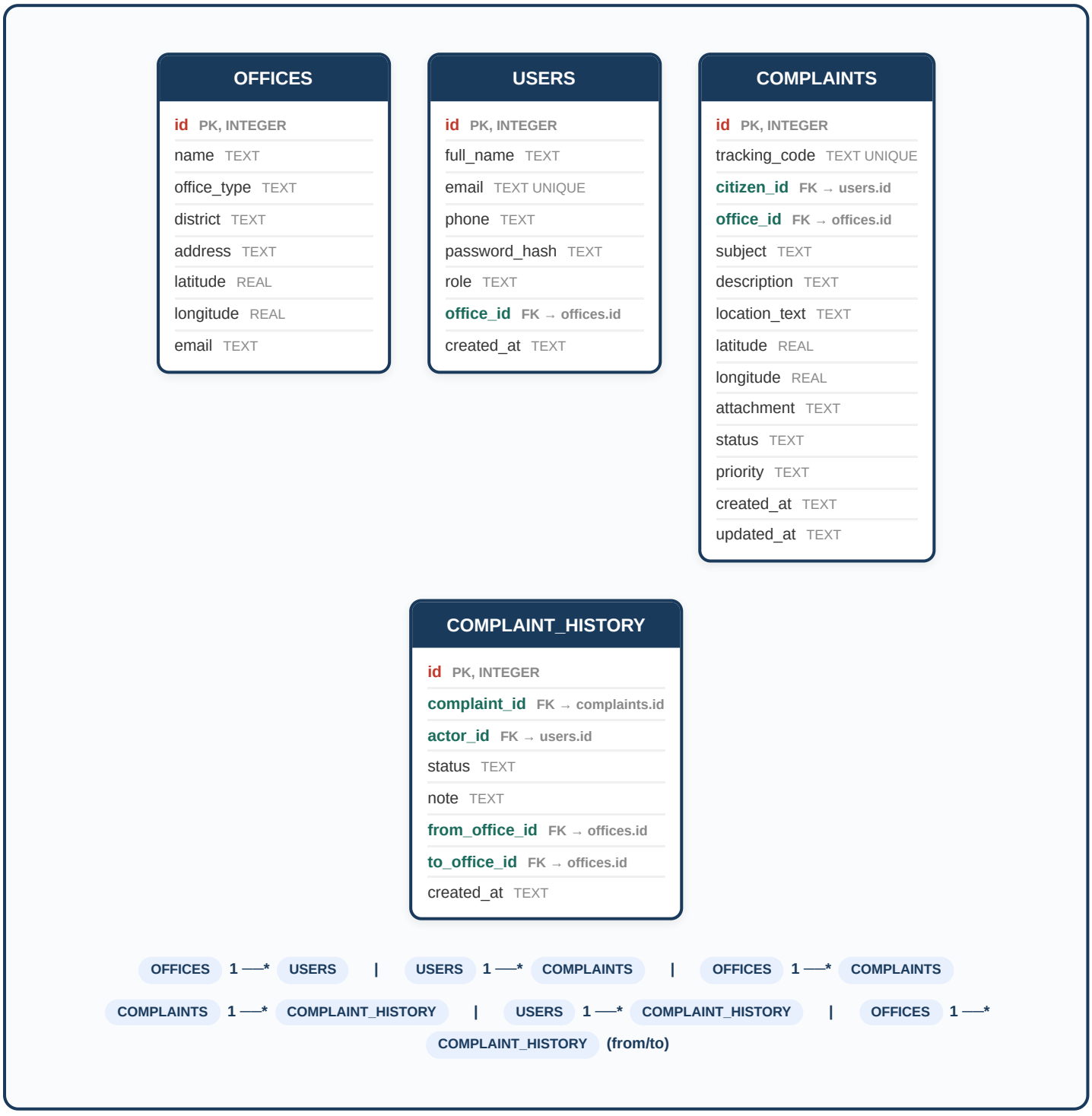
PARATHI — Database Design

Public Complaint Management System — DBMS Course Project | SQLite Database

Primary Key (PK) Foreign Key (FK)

1. Entity-Relationship Diagram



2. Foreign Key Relationships

Child Table	Foreign Key	Parent Table	Parent PK
users	office_id	offices	id
complaints	citizen_id	users	id

complaints	**office_id**	offices	id
complaint_history	**complaint_id**	complaints	id
complaint_history	**actor_id**	users	id
complaint_history	**from_office_id**	offices	id
complaint_history	**to_office_id**	offices	id

3. Sample Queries

```
SELECT c.*, o.name AS office_name
FROM complaints c JOIN offices o ON c.office_id = o.id
WHERE c.office_id = ?;
```

```
SELECT status, COUNT(*) AS count FROM complaints
WHERE office_id = ? GROUP BY status;
```

```
SELECT h.*, u.full_name AS actor_name
FROM complaint_history h JOIN users u ON h.actor_id = u.id
WHERE h.complaint_id = ? ORDER BY h.created_at;
```

4. Normalization

- **1NF:** Atomic attributes, no repeating groups. History in separate table.
- **2NF:** No partial dependencies on composite keys.
- **3NF:** No transitive dependencies. Office details stored once, referenced by FK.

5. Seed Data

5 offices, 6 users (1 citizen + 5 officers), 3 sample complaints. All seeded on first run.

6. Access Control

Officers only see complaints where `complaints.office_id = users.office_id`. Enforced at SQL level, not just UI.